Different Normal Forms

First Normal Form (1NF) >>

- All the columns/attributes are single valued.
- Primary key exists.
- Functional Dependencies are identified.

Second Normal Form (2NF) >>

- It is in 1NF.
- No partial dependency exists; that is no non-prime attribute is dependent on the proper subset of any candidate key of table.
- Prime attribute: an attribute that is a part of any candidate key is known as prime attribute.
- For each non-trivial FD of the form **X → Y**
  - Either Y is a prime/key attribute.
  - If Y is non-prime then X is **not a proper subset** of any candidate key of the table. That is either fully dependent on the candidate key or fully dependent on other non-prime attributes.
- Example:
  - Given, R = (A, B, C, D, E, F, G) and F = {A→B, C→DEF, E→F, AC → BDEFG }
  - Here, Candidate Key={ AC }, proper subset={ {}, A, C }, non-prime attributes = { B, D, E, F, G }
  - But in A → B, B is non-prime and A is a proper subset of candidate key
  - So it is not in 2NF

Third Normal Form (3NF) >>

- It is in 2NF.
- No transitive dependency exists.
- For each non-trivial FD of the form **X → Y**
  - Either Y is a prime/key attribute
  - If Y is non-prime then X is a superkey of the table.
- Example:
  - If R=(C, D, E, F) and F={ C → DEF, E → F }
  - Here, candidate key = { C }, proper subset = {φ}, non-prime attributes = {D, E, F}
  - In C → DEF; D,E,F each is non-prime and C is not proper subset of candidate key and it is a superkey and in E → F; F is non-prime and E is also not a proper subset of candidate keys but not superkey.
  - So, it is in 2NF but not in 3NF

Boyce-Codd Normal Form (BCNF/3.5NF) >>

- It is in 3NF.
- For each no trivial FD of the form **X → Y**
  - X is the superkey of the table.
- If we do not have redundancy in **F**, then for each **X → Y** of $\mathbf{F_C}$, X must be a candidate key.
- Example:
  - If R = (A, B, C, D) and F = { AB → CD, C → B }
  - Here, Candidate Keys = { AB, AC }
  - $(AB)^+$ = ABCD i.e. superkey and $(C)^+$ = CB not a superkey but B is a key/prime attribute
  - So, it is in 3NF but not in BCNF(as C is not a superkey)

Conversion to BCNF form >>

Properties:

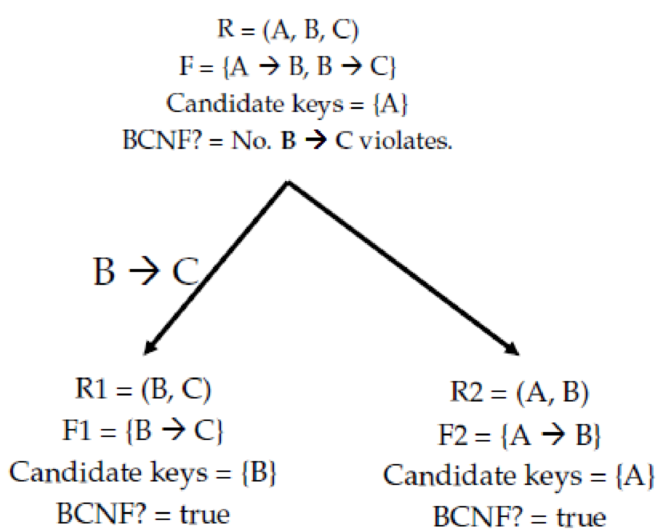
- No redundancy
- Lossless-join
- But may/may not preserve dependency

Steps to follow:

Given, A Relational Schema R and set of Functional Dependencies F

- *Step 1:* Find the minimal cover of F, F_c
- *Step 2:* Find a dependency in F_c that violates the BCNF form(that is left side is not a superkey), suppose $X \rightarrow Y$
- *Step 3:* Divide R into two Relational Schema ($R_1 = (X)^+$, F_1)and ($R_2 = R - (X)^+ \cup \{X\}$, F_2)
- *Step 4:* for each FD in F_C (let, $A \rightarrow B$),
  -> if all the attributes of A and B fall under R_1 then $A \rightarrow B$ is the member of F_1
  -> if all the attributes of A and B fall under R_2, then $A \rightarrow B$ is the member of F_2
  -> if the FD doesn't fall under R_1 or R_2 then replace each attribute of X^+ with X in $A \rightarrow B$ and recheck if still now it doesn't fall for R_1 and R_2 then remove this dependency (i.e. not preserving dependency)
- *Step 5:* repeat steps 1-5 for R_1 and R_2

Example 1: here, F= F_c



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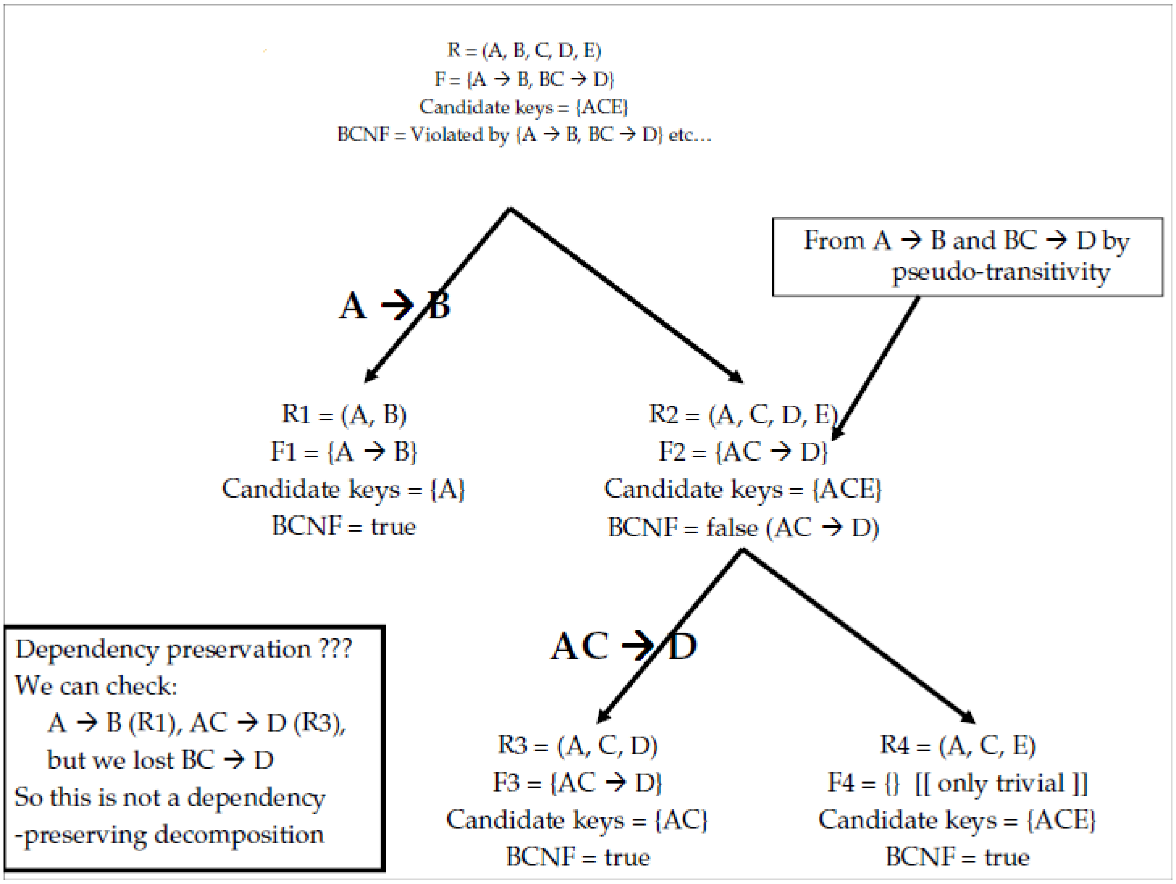
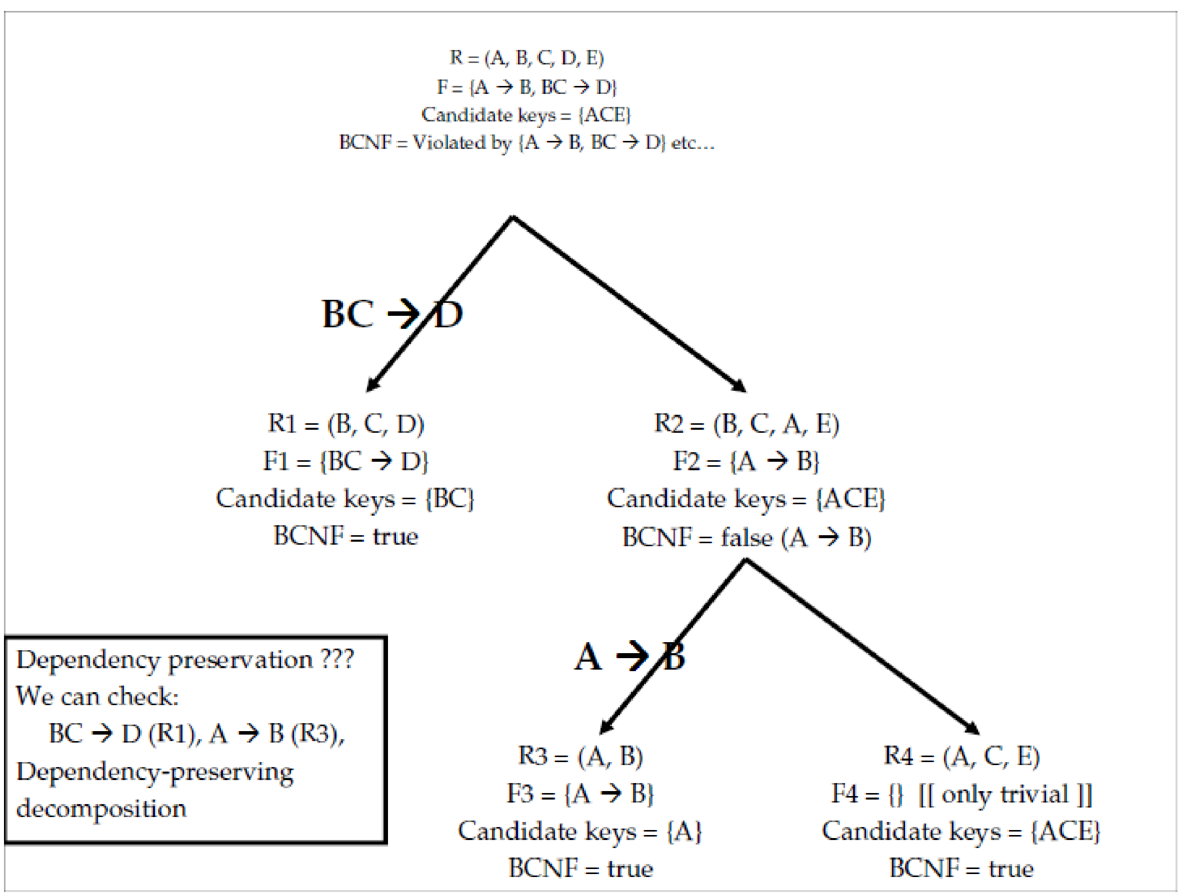
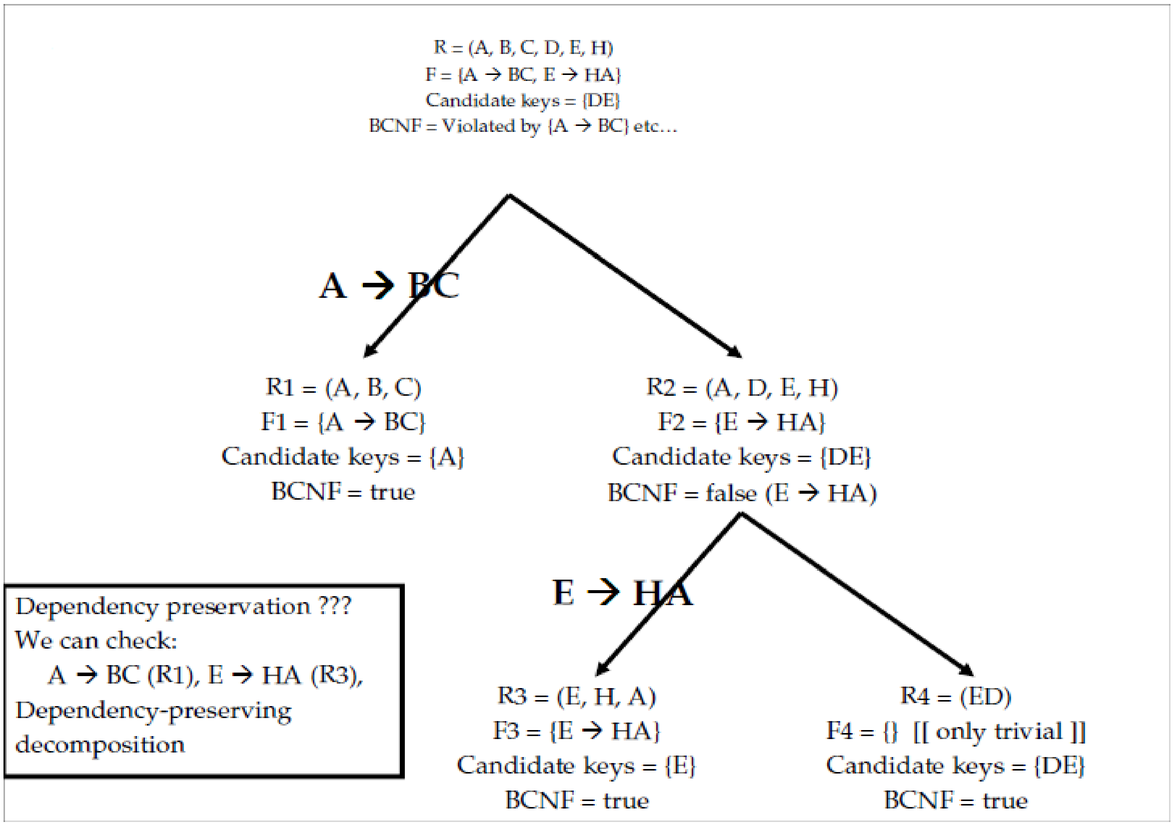
Example 2: here, F= F_c



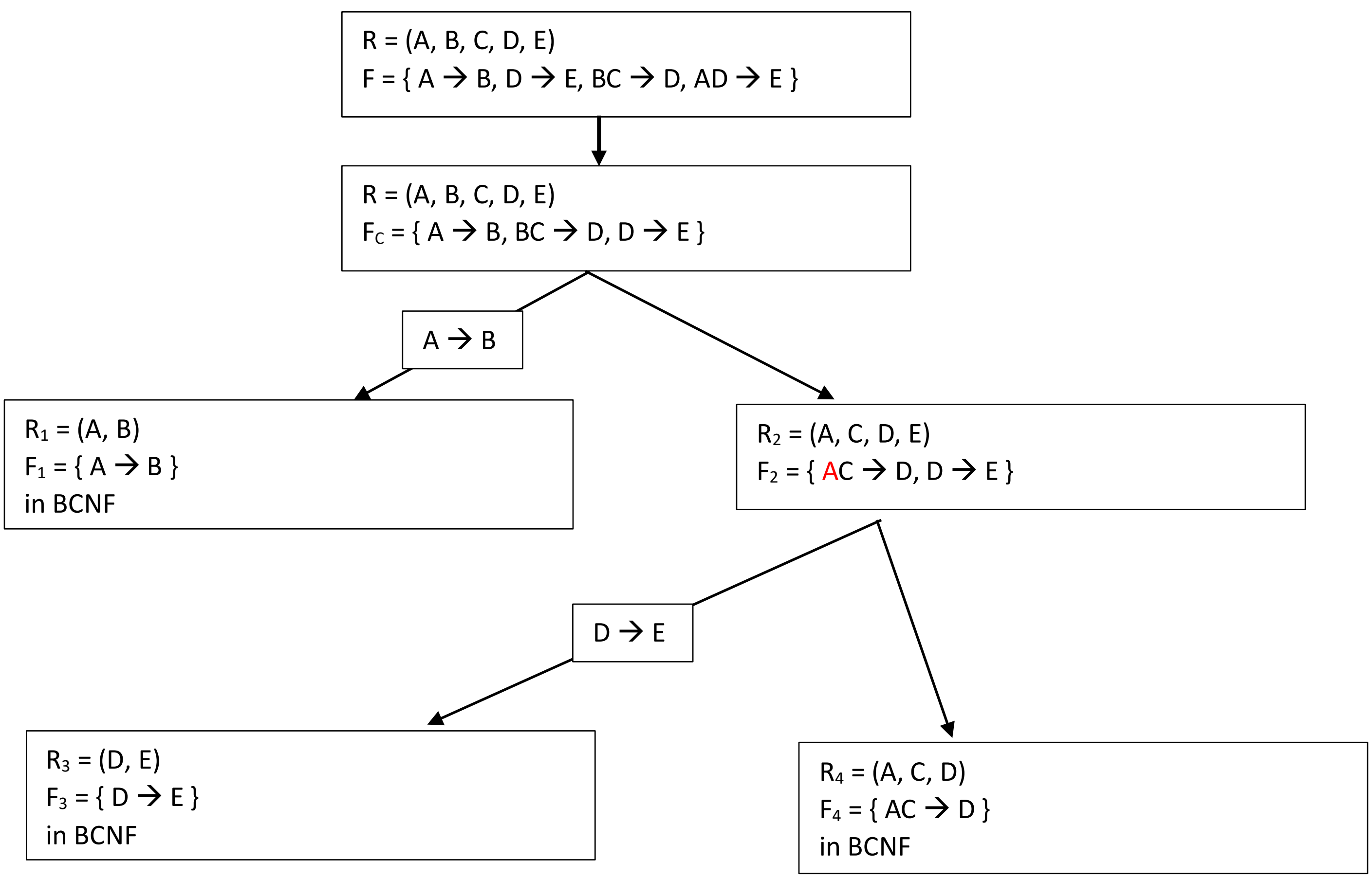
Alternate version:



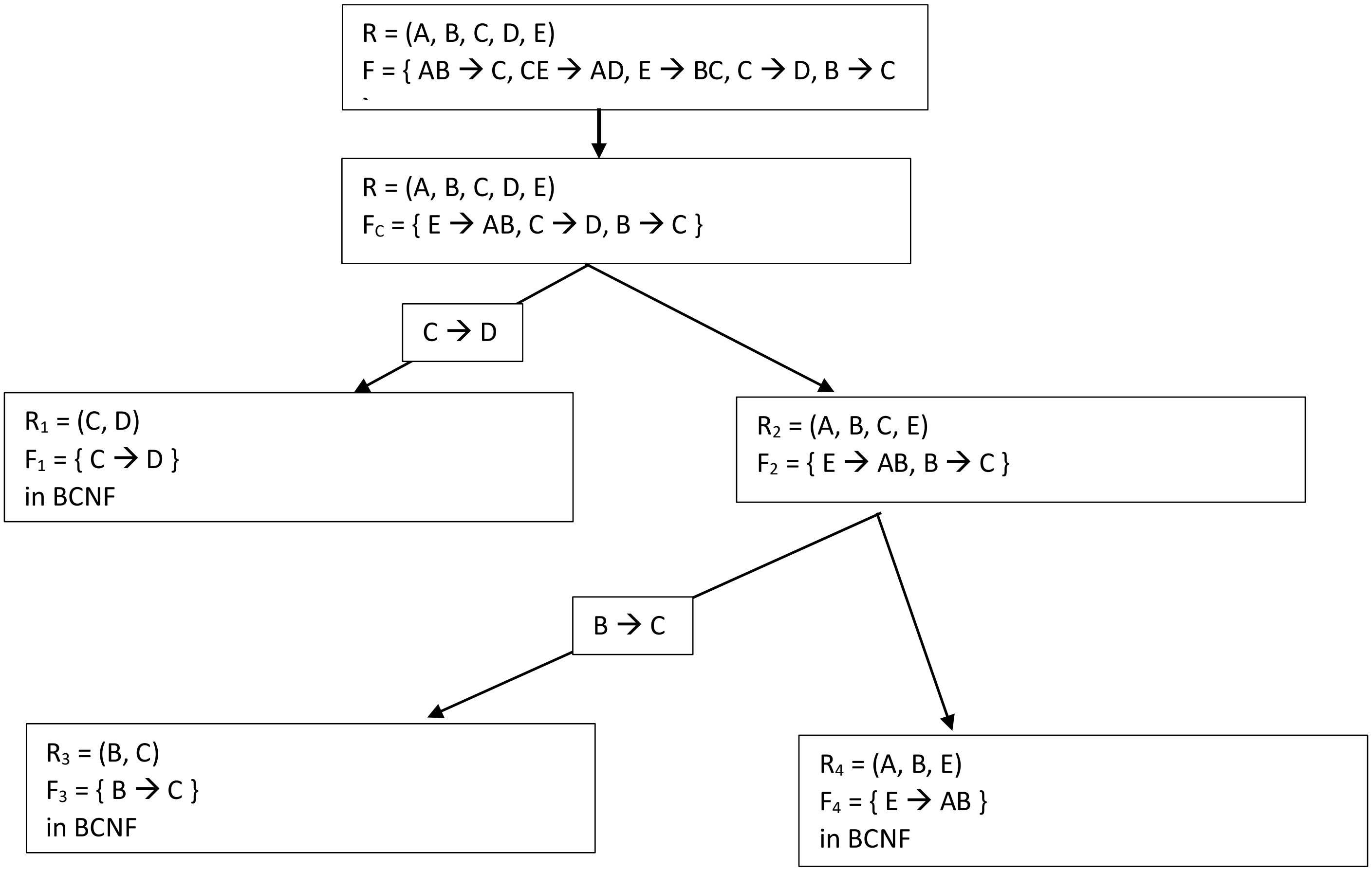
Example 3: here, $F = F_c$



Example 4:



Example 5:



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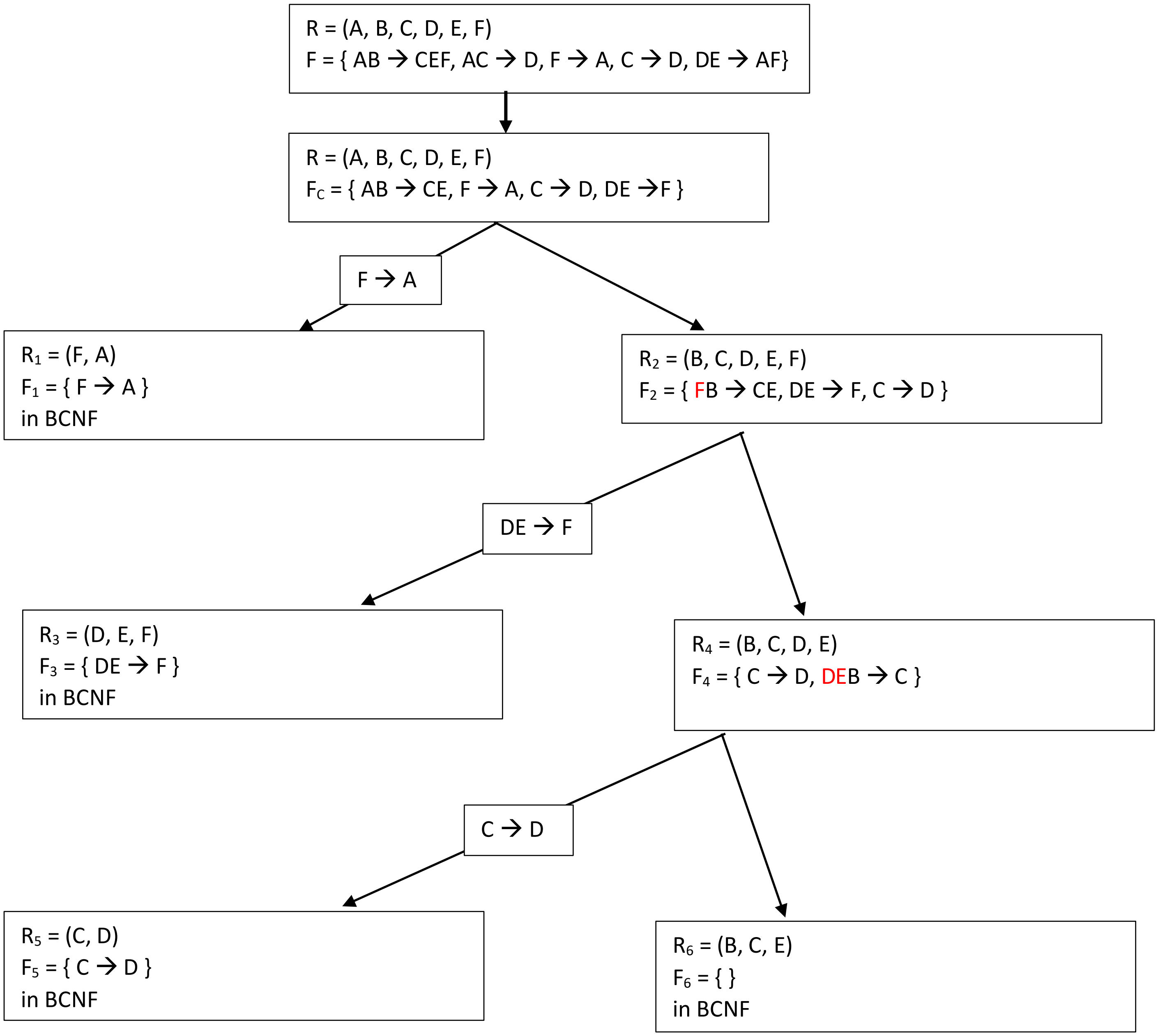
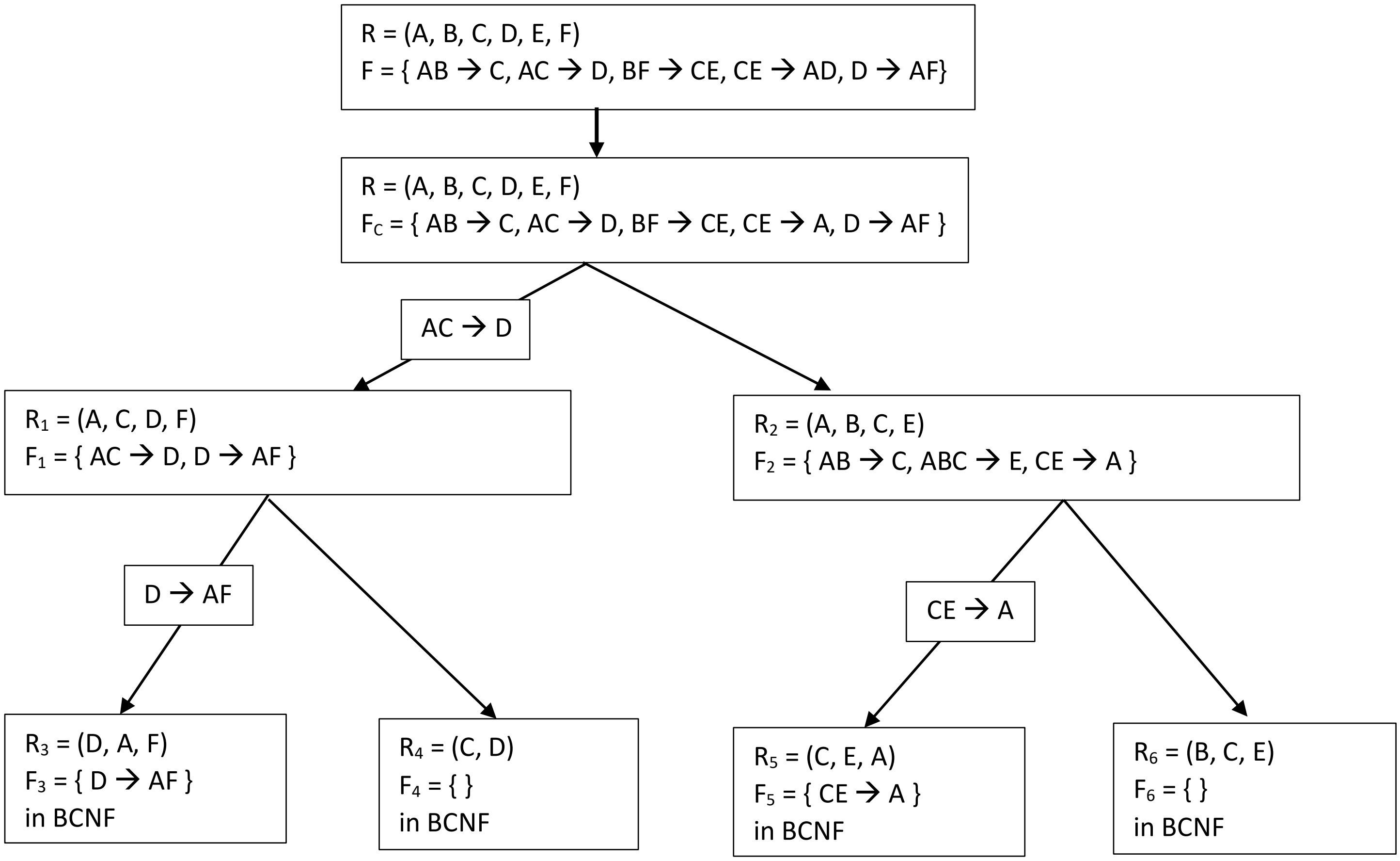
Example 6:


R = (A, B, C, D, E, F)
F = { AB → CEF, AC → D, F → A, C → D, DE → AF}
R = (A, B, C, D, E, F)
F_C = { AB → CE, F → A, C → D, DE → F }
F → A
R_1 = (F, A)
F_1 = { F → A }
in BCNF
R_2 = (B, C, D, E, F)
F_2 = { FB → CE, DE → F, C → D }
DE → F
R_3 = (D, E, F)
F_3 = { DE → F }
in BCNF
R_4 = (B, C, D, E)
F_4 = { C → D, DEB → C }
C → D
R_5 = (C, D)
F_5 = { C → D }
in BCNF
R_6 = (B, C, E)
F_6 = { }
in BCNF

Example 7:



Practice 1:

R = (A, B, C, D, E), F = { AB $\rightarrow$ CE, AC $\rightarrow$ D, BE $\rightarrow$ CD, C$\rightarrow$D, E $\rightarrow$ AB }

Final Result >>

R1=(C,D) And R2 = (A,B,C,E)

Practice 2:

R = (A, B, C, D, E, F, G, H, I), F = {A $\rightarrow$ B, E $\rightarrow$ FG, G $\rightarrow$ H, AEC $\rightarrow$ I, AC $\rightarrow$ D, D$\rightarrow$ C}

Final Result >>

R1=(A, B), R5=(G, H), R6=(E, G, F), R8=(A, C, E, I), R9=(D, C), R10=(A, D)